

From powerful AI models to trustworthy scientific AI systems

Scientific discovery will require more than isolated AI models. This PhD project will study how models, data, tools, domain knowledge, uncertainty, and human expertise can be integrated into AI systems that are reliable, interpretable, adaptable, and useful for science.

We are looking for a highly motivated PhD candidate to join the Institute for Explainable Machine Learning at Helmholtz Munich and the Chair of Interpretable and Reliable Machine Learning at the Technical University of Munich.

Project Description

Modern AI research has produced highly capable models, but scientific workflows require systems that can combine heterogeneous data, multiple models, external tools, domain constraints, and expert feedback while remaining transparent, uncertainty-aware, and robust. This PhD project will study how assumptions, limitations, uncertainty, errors, and provenance propagate across integrated AI systems, and how such systems can be reliably updated, inspected, and evaluated. Research areas include:

- **Explainable AI and Mechanistic Interpretability** of unimodal and multimodal foundation models.
- **Multimodal Alignment and Representation Learning** for foundation models, including vision-language models (VLMs), multimodal foundation models, and multimodal reasoning.
- **Reliable Adaptation and Continual Learning** for AI systems, including fine-tuning and parameter-efficient adaptation.
- **AI for Science**, including agentic scientific workflows, uncertainty estimation, benchmarking, and applications to biological and medical data.

The exact project will be shaped together with the selected candidate. The PhD student will join a collaborative research environment across Helmholtz Munich, TUM, and an international network of collaborators.

Requirements

- A master's degree, or equivalent, in computer science, machine learning, mathematics, statistics, physics, engineering, or a related field
- A strong foundation in machine learning (prior publications in top-tier ML/CV/NLP conferences is a plus)
- Strong programming skills and experience with modern machine learning frameworks
- Interest in explainability, reliability, multimodal learning, foundation models, AI agents, or AI for science
- Good communication skills and enthusiasm for interdisciplinary research
- The ability to work independently while contributing to a collaborative research environment

We especially encourage applications from candidates who enjoy both conceptual thinking and technical implementation, and who are excited about shaping a new research direction.

Application / Contact

Please send your application with the subject line containing **[PhD 2026 EML]** to job-applications-eml@helmholtz-munich.de as a **single PDF file** containing:

- CV.
- Transcripts and degree certificates.
- A short research statement describing your interests, relevant background, and how they relate to our team / this PhD project.
- Contact information for two references.

Priority deadline: 10 August 2026, 23:59 CET.

Applications received by this date will receive full consideration. Applications may continue to be reviewed until the position is filled.